IGCSE 0607 Practice Questions — E1.14 — Combined Questions & Markscheme

Time

Total Questions: 9 — **Total Marks:** 44 Marks

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  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 A train timetable shows that a train leaves Paris at 09 55 and arrives in Lyon at 12 10.

(a) Work out the time taken for this journey in hours and minutes.

hours ..... , minutes ..... **[1]**

(b) Another train leaves Paris at 16 45.
This journey takes 2 hours and 35 minutes.
Work out the time this train arrives in Lyon.

..... **[2]**

2 A flight departs from London to Dubai at 21 : 45 local time.

The flight time is 6 hours and 50 minutes.
The local time in Dubai is 4 hours ahead of the local time in London.
Work out the local time in Dubai when the flight arrives.

..... **[3]**

3 A flight departs from Dubai at 14 30 local time and arrives in Sydney.
The flight time is 14 hours and 15 minutes.
The local time in Sydney is 7 hours ahead of the local time in Dubai.
Work out the local time in Sydney when the flight arrives.

..... **[3]**

4 A train travels from Station A to Station B. The total distance is 420 km.
The train departs Station A at 08 : 15 and arrives at Station B at 11 : 03.

(a) Calculate the time taken for the journey, in hours and minutes.

hours, minutes **[2]**

(b) Calculate the average speed of the train in km/h.

..... km/h **[3]**

5 A ship travels a distance of 920 km at an average speed of 24 km/h.

(a) Work out the time taken for the journey.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

(b) The ship left the port on Wednesday at 17 50.
Determine the day and the time it arrives at its destination.

Day ..... , Time ..... **[4]**

6 Consider the following applications of time, speed, and dates.

(a) A satellite completes one orbit of the Earth in exactly 92 minutes.
Calculate the number of full orbits it completes in exactly one non-leap year (365 days).

..... orbits **[4]**

(b) A plane flies from City P to City Q. The distance is 11 440 km and the average speed is 880 km/h.

(i) Calculate the flight time.

..... hours **[1]**

(ii) City P is in the time zone UTC−4.
City Q is in the time zone UTC+9.
The plane departs from City P on Monday at 22 : 45 local time.
Find the local day and time of arrival in City Q.

Day = ... , Time = ... **[3]**

7 A flight leaves Dubai at 14 30 local time on Friday and arrives in London at 19 15 local time on the same day.
The local time in Dubai is 4 hours ahead of the local time in London.

(a) Work out the duration of the flight in hours and minutes.

hours, minutes **[2]**

(b) A second flight leaves Dubai at 22 45 local time on Friday.
The duration of this flight is 7 hours and 20 minutes.
Work out the local time and the day it arrives in London.

Time = ... , Day = ... **[2]**

8 Light travels at 3×10^5 km/s.

(a) Calculate the distance light travels in one year (365 days).
Give your answer in standard form.

..... km **[2]**

(b) The distance from Earth to a star is 4.5×10^{13} km.
Calculate the time taken for light to travel from the star to Earth.
Give your answer in years, correct to 3 significant figures.

..... years **[2]**

9 An aircraft flies from City A to City B, a distance of 6600 km.
The local time at City B is 5 hours behind the local time at City A.
The aircraft leaves City A at 21 40 local time on Tuesday.
The average speed of the aircraft is 800 km/h.

(a) Calculate the time of the flight in hours and minutes.

hours, minutes **[2]**

(b) Find the local time and the day when the aircraft arrives at City B.

Time = ... , Day = ... **[2]**

(c) The aircraft stays at City B for 14 hours and 20 minutes before flying back to City A.
The return flight takes 7 hours and 45 minutes.
Find the local time and the day when the aircraft arrives back at City A.

Time = ... , Day = ... **[3]**

1 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 Arrival time in London local time = 21 : 45 + 6 hours 50 minutes **[M1]**

04 : 35 (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** 11 : 03 − 08 : 15 **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \frac{48}{60}$ hours = 0.8 hours **[M1]**

Average speed $= \frac{420}{2.8}$ **[M1]**

150 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** 920 ÷ 24 **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** Minutes in a year = 365 × 24 × 60 **[M1]**

525600 minutes **[M1]**

Number of orbits $= \frac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) (i) $\frac{11440}{880} = 13$ **[B1]**

(ii) Time difference $= 9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time = 22 : 45 (Monday) +13 hours = 11 : 45 (Tuesday) **[M1]**

Arrival time in City Q's local time = 11 : 45 (Tuesday) +13 hours (flight time) = 00 : 45 (Wednesday) **[A1]**

[Back to Question] *[Total Marks: 8]*

7 (a) London departure time = 10 30 or Dubai arrival time = 23 15 **[M1]**

8 hours 45 minutes **[A1]**

(b) London departure time = 18 45 or Dubai arrival time = 06 05 (Saturday) **[M1]**

02 05 Saturday **[A1]**

[Back to Question] *[Total Marks: 4]*

8 (a) $3 \times 10^5 \times 60 \times 60 \times 24 \times 365$ **[M1]**

9.46×10^{12} or 9.4608×10^{12} **[A1]**

(b) $\frac{4.5 \times 10^{13}}{9.4608 \times 10^{12}}$ **[M1]**

4.76 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 (a) $\frac{6600}{800}$ or 8.25 **[M1]**

8 hours 15 minutes **[A1]**

(b) Arrival time in City A time = 05 55 **[M1]**

00 55 Wednesday **[A1]**

(c) Departure from City B = 15 15 (local time) or 20 15 (City A time) **[M1]**

Arrival in City B time = 23 00 **[M1]**

04 00 Thursday **[A1]**

[Back to Question] *[Total Marks: 7]*